

CarterPCs — fix-v14

Section 04 / Hardware: visual and responsive QA pass

Six defects found, six fixed. Four of them were one defect wearing different clothes: the section was 210-547px taller than the viewport it pins in.

Baseline d0b8ecf · 3 files changed · 262 insertions, 20 deletions
typecheck 0 · lint 0 · unit 42/42 · build pass · e2e 69/69 · git diff --check clean
Nothing committed, pushed, reset, cleaned or reverted.

Executive summary

The desktop trigger pins the section for 0.65 of a viewport. Across 1024/1440/1920 the tags line was **off-screen for the pin's entire duration**, and at 1024 the third beat was too: the reader scrolled ~585px, nothing moved, and the part they had not read yet stayed below the fold. Meanwhile there were 160-179px of dead space above the first line.

The copy column and the object deliberately shared a grid column. That works until a headline is long enough to reach it: at 1440 the rear plate crossed **4,856px² of live headline ink in English and 6,229px² in Spanish**, greying out the last glyph of “the”.

And the #hardware anchor landed **101px behind the fixed bar** at 390px.

Fixed by re-flowing the desktop grid (copy and object in their own columns, the three beats as a band across the base), capping the object on small screens, hanging the pin off the section's bottom edge, and giving the object enough material contrast to be seen at all. The section is now exactly one viewport at 1024x768, 1440x900 and 1920x1080, with zero collisions in all three languages. No copy, translation, data or interaction changed.

Every “before” number was re-measured by building the baseline commit's Hardware files into a second app and serving it alongside the working tree, then running the same probe against both.

Files changed

File	Lines	What
src/sections/hardware/Hardware.module.css	+120 / -20	responsive anchor offset; object capped and outboard on small screens; desktop grid re-flow; object material
src/sections/hardware/Hardware.tsx	+14 / -1	pin trigger moved from the section's top edge to its bottom edge
e2e/app.spec.ts	+127	two regression tests and two helpers

D1 — the anchor landed behind the fixed bar

The bar is 5rem tall only from 1024px up; below that its link row wraps to two lines, then three. Where it wraps is content-driven, so it moves with the language — English is back to one line at 480px while French still needs two. One scroll-margin could not cover that.

Viewport	Bar height	Before	After
390×844	197.0	− 101.4px (behind the bar)	+10.6px
768×1024	127.7	− 31.5px (behind the bar)	+16.5px
1024×768	80.0	+16.4px	+16.4px
1440×900	80.0	+15.8px	+15.8px
1920×1080	89.3	+18.0px	+18.0px

D2 — the object crossed the headline

Measured against the PAINTED planes, not the figure's box:

Viewport	Before (EN)	Before (ES)	After (all languages)
1024×768	167px ² (support copy)	535px ² (support copy)	0
1440×900	4,856px ² (headline)	6,229px ² (headline)	0
1920×1080	0	0	0

Copy now takes columns 1-6, the object 7-12, with a full grid gap between. Gutter between the headline's longest ink and the object's left edge, after: 79px at 1024, 166px at 1440, 60px at 1920 — in every language.

D3 — the pin held a section that did not fit

Measurement	Before	After
section height @1024×768	1,220-1,315px	768px
section height @1440×900	1,110-1,137px	900px
section height @1920×1080	1,243px	1,080px
tags visible during the pin	no , at all three sizes	yes , at all three
third beat visible during the pin @1024	no	yes

The height came out of the beats: a vertical stack in the left column was ~390px, the same three beats across the base are 119-146px. Together with tighter block padding (9rem→7rem top, 6rem→4rem bottom) and a row gap that was silently inheriting the mobile rule's 4rem, that is the 210-547px.

The trigger also moved from the section's top edge to its bottom edge. When the section fits the viewport the two are the same scroll position to the pixel; they differ only when it does not — a short window, a large zoom. Tested at 1280x720, 1366x630, 1024x600, 1440x820 and 2560x900: the object, the last beat and the tags all become fully visible while scrolling, and the pin engages only afterwards.

```
start: 'bottom bottom', // was 'top top'
end:   () => `${Math.round(window.innerHeight * 0.65)}`,
```

D4 — dead space above, cut-off below

Space between the bar and the first line after an anchor jump: 160.4 → 112.4px (1024), 159.8 → 111.8px (1440), 179.3 → 125.5px (1920).

D5 — the primary visual was a void

The surface token on the base token is **1.16:1**, and the old gradient diluted it with transparency and ran it back to the base colour by 82%: the largest element in the section carried the least information in it. The lit face is now a mix of the TEXT token into the surface token (~1.6:1), with a 1px inset edge describing the silhouette, a brighter lit top, and a mask that fades the trailing corner over 18% instead of 32% — the old ramp erased the corner cut and the edge with it. One rule, and it inverts correctly on the light theme.

The environmental “04” also became legible as a consequence: it was 8% of the text token sitting behind the object and below the fold. Its style is unchanged — 8% matches Closing's numeral. Beat text over it measures 6.5:1 (dark) and 4.88:1 (light), both AA.

D6 — the object swallowed tablet and mobile landscape

Viewport	Object before	Object after	Section height
768×1024	706×883	384×480	1,893 → 1,490
844×390	776×971	187×234	1,990 → 1,254
390×844	350×438	350×438 (unchanged)	1,444 (unchanged)
320×568	280×350	273×341	1,410 (unchanged)

From 600px up — where 322px of column was left over — the object moves to the outer edge instead of hugging the copy's left margin, rehearsing the desktop composition one breakpoint early rather than inventing a third arrangement.

Verified unchanged

Check	Result
copy, translations, data	untouched — no i18n or data file in the diff
contrast, 8 text roles × 2 themes	identical before and after: dark 7.69-17.8, light 5.16-16.7, all ≥ AA
interactions	the section has 0 focusable elements ; tap-target sizing is not applicable
entrance reveal	9 revealed elements, 0 left faded, clipped or shifted (EN/FR, mobile and desktop)
depth-separation motion	amplitudes identical (rear −16,−55 @0.92; front +18,+60 @1.06; numeral −14) — and the full separate-and-recompose arc now plays inside the pin, where before only the recompose half did
reduced motion	nothing hidden, clipped or transformed; section 900px; overflow 0
system theme	resolves in both OS settings; accents match the stored-theme values
horizontal overflow	0 across 42 viewport × language × theme combinations
clipped text	0, in all three languages at all seven viewports
sections above	Hardware's own top is unmoved: 5,398.8 @1440 and 7,116.6 @390, before and after
section below	Content Universe moves up 210px at 1440 — entirely because Hardware is 210px shorter. Both seams still meet at a 0.0px gap

Remaining design decisions

- 1. 2560×900.** The root font-size scales to 21.125px at that width, so the section comes to 1,226px and cannot fit a 900px viewport. Nothing is stranded — the pin waits — but the object is never framed whole. Outside the required matrix; stated rather than special-cased.
- 2. After an anchor jump the last ~96px sits below the fold** (the tags line at 1440), because a 100vh section landing below a fixed bar cannot show all of itself. One scroll later the pin frames it exactly. Inherent, not fixed.
- 3. Tablet keeps the stacked order.** A two-column split at 768 would put a 70px headline in a ~337px column and rewrap every headline in three languages; the object moving outboard gets most of the benefit for none of that risk.
- 4. The object is still a placeholder.** No approved CarterPCs hardware media exists, and the brief rules out inventing any. The media layer remains the single drop-in target; everything here improves how it reads as material, not what it depicts.

04 / HARDWARE

HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not
what the box promises.

01 BUILD

The machine as a finished object —
assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined
up close: GPU, CPU, cooling, storage.

01a-before-1440x900-en-dark.png

BEFORE — 1440x900 EN dark. The rear plate crosses the headline; the tail of the section is below the fold.

04 / HARDWARE

HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not what the box promises.

01 BUILD

The machine as a finished object — assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined up close: GPU, CPU, cooling, storage.

03 PERFORMANCE

Not a benchmark chart — what the hardware actually enables day to day.

01b-after-1440x900-en-dark.png

AFTER — 1440x900 EN dark. Copy and object in their own columns, the three beats as a band, the whole section in one viewport.

04 / HARDWARE ————— HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not what the box promises.

01 BUILD

The machine as a finished object — assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined up close: GPU, CPU, cooling, storage.

03 PERFORMANCE

02a-before-1920x1080-en-dark.png

BEFORE — 1920x1080 EN dark. The object occupies 1,021x851 of near-black on near-black.

04 / HARDWARE

HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not what the box promises.

01 BUILD

The machine as a finished object — assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined up close: GPU, CPU, cooling, storage.

03 PERFORMANCE

Not a benchmark chart — what the hardware actually enables day to day.

02b-after-1920x1080-en-dark.png

AFTER — 1920x1080 EN dark. The object reads as a lit surface; the environmental 04 is legible for the first time.

04 /
HARDWARE

EXPERIENCIA DE
HARDWARE

DENTRO DEL MONTAJE

Construido desde dentro.

Cada componente juzgado por lo que hace,
no por lo que promete la caja.

01 MONTAJE

La máquina como objeto

03a-before-1024x768-es-dark.png

BEFORE — 1024x768 ES dark. Section 1,315px in a 768px viewport; the third beat and the tags are unreachable during the pin.

04 / HARDWARE

EXPERIENCIA DE HARDWARE

DENTRO DEL MONTAJE

Construido desde dentro.

Cada componente juzgado por lo que hace,
no por lo que promete la caja.



01 MONTAJE

La máquina como objeto
acabado: montada, cableada y
encendida.

02 COMPONENTES

Las piezas que de verdad
importan, examinadas de cerca:
GPU, CPU, refrigeración,
almacenamiento.

03 RENDIMIENTO

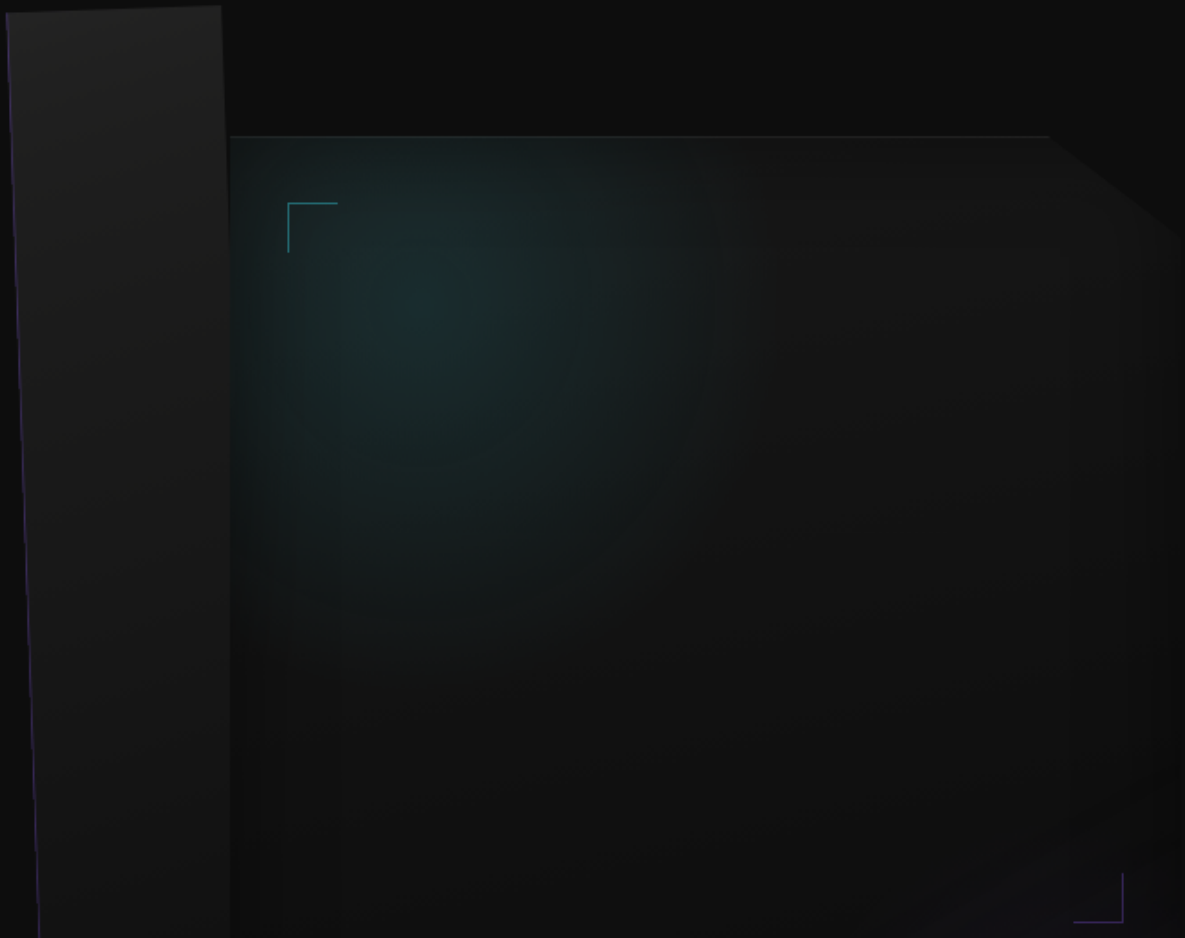
No es una tabla de benchmarks:
lo que el hardware permite de
verdad cada día.

03b-after-1024x768-es-dark.png

AFTER — 1024x768 ES dark. Exactly 768px: everything the pin holds is visible.

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il fait, pas
sur ce que promet la boîte.

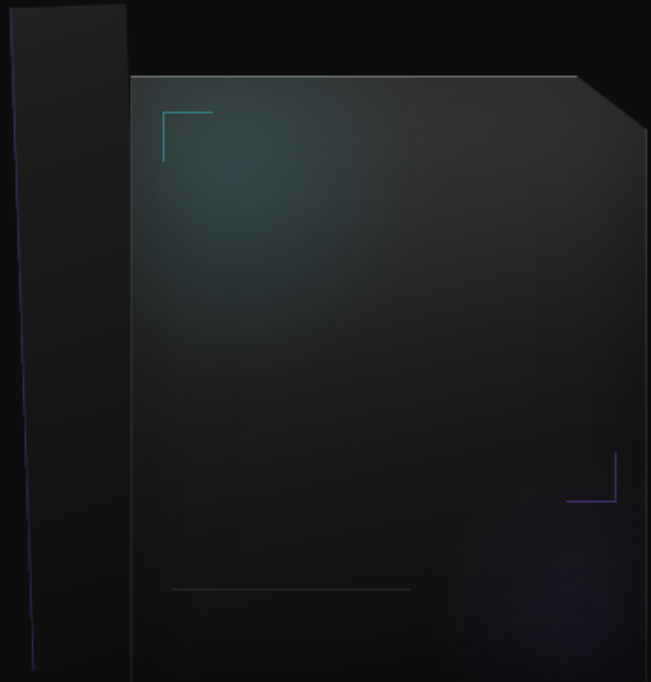




AU CŒUR DU MONTAGE

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il fait, pas sur ce que promet la boîte.





INSIDE THE BUILD

Built from the inside out

05a-before-844x390-en-dark.png

BEFORE — 844x390 EN dark, mobile landscape. The object was 776x971 — two and a half viewport heights.

SHARE

WATCH ON YOUTUBE

* CarterPCs •

About Carter ↗



Work

Systems

Process

Impact

Content

Universe

04 / HARDWARE

HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the

05b-after-844x390-en-dark.png

AFTER — 844x390 EN dark. 187x234, and the section is 1,254px instead of 1,990px.

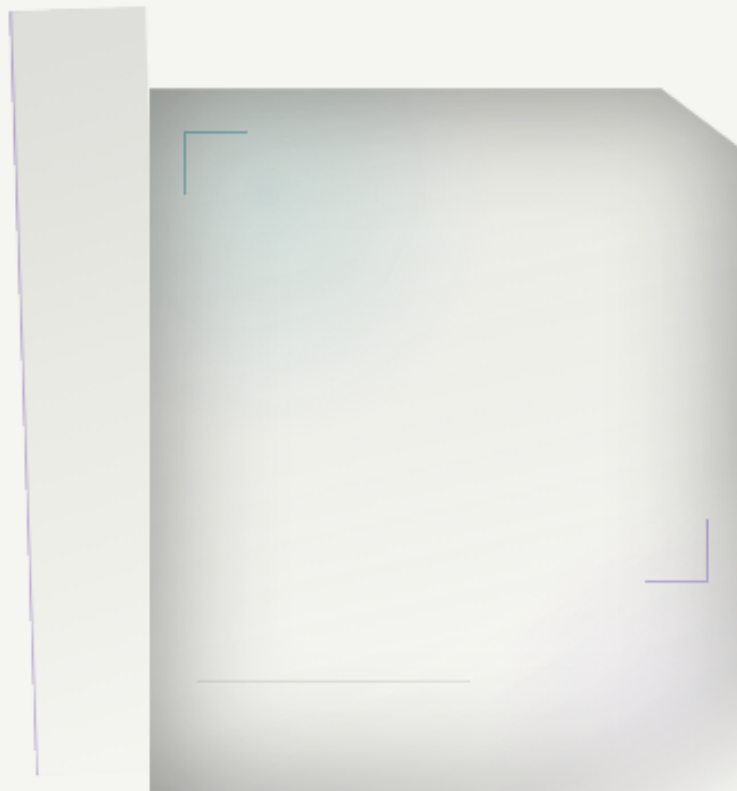
Work Systems Process Impact Content
Universe

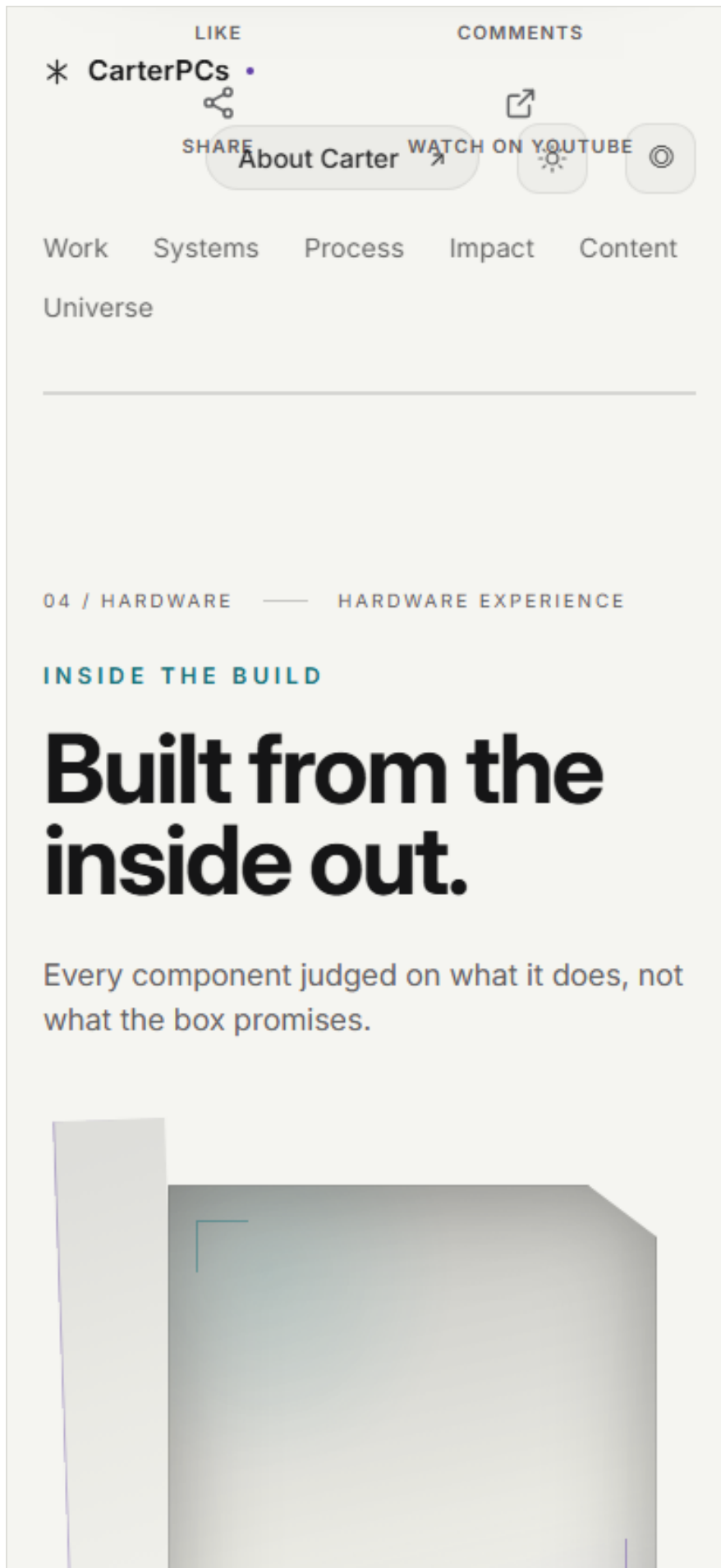
04 / HARDWARE — HARDWARE EXPERIENCE

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not
what the box promises.





06b-after-390x844-en-light.png

AFTER — 390x844 EN light. The section lands 10.6px below the bar.

* CarterPCs •

À propos de Carter ➤



Projets

Systèmes

Processus

Impact

Contenu

Univers

04 /
MATÉRIEL

EXPÉRIENCE
MATÉRIELLE

AU CŒUR DU MONTAGE

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il
fait, pas sur ce que promet la boîte.

J'AIME

COMMENTAIRES

* CarterPCs •



PARTAGER
À propos de Carter

VOIR SUR YOUTUBE



Projets

Systèmes

Processus

Impact

Contenu

Univers

04 /
MATÉRIEL

EXPÉRIENCE
MATÉRIELLE

AU CŒUR DU MONTAGE

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il
fait, pas sur ce que promet la boîte.

04 / MATÉRIEL ————— EXPÉRIENCE MATÉRIELLE

AU CŒUR DU MONTAGE

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il fait, pas sur ce que promet la boîte.

01 MONTAGE

La machine comme objet fini — assemblée, câblée et sous tension.

02 COMPOSANTS

Les pièces qui comptent vraiment, examinées de près : GPU, CPU, refroidissement, stockage.

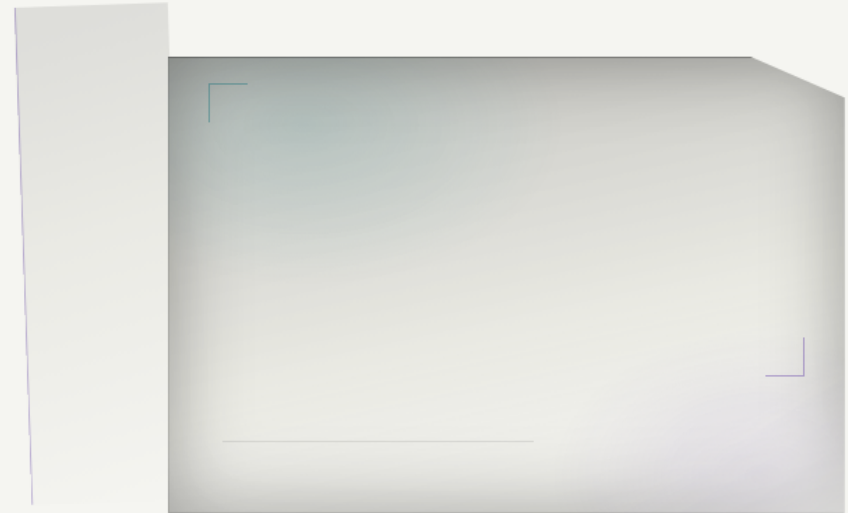


08a-before-1440x900-fr-light.png

BEFORE — 1440x900 FR light. The plate crosses the headline here too, in the light theme.

Conçu de l'intérieur.

Chaque composant jugé sur ce qu'il fait, pas sur ce que promet la boîte.



01 MONTAGE

La machine comme objet fini — assemblée, câblée et sous tension.

02 COMPOSANTS

Les pièces qui comptent vraiment, examinées de près : GPU, CPU, refroidissement, stockage.

03 PERFORMANCE

Pas un graphique de benchmark — ce que le matériel permet vraiment au quotidien.

04 / HARDWARE

EXPERIENCIA DE HARDWARE

DENTRO DEL MONTAJE

Construido desde dentro.

Cada componente juzgado por lo que hace,
no por lo que promete la caja.

01 MONTAJE

La máquina como objeto acabado: montada, cableada y encendida.

02 COMPONENTES

Las piezas que de verdad importan, examinadas de cerca: GPU, CPU, refrigeración, almacenamiento.

03 RENDIMIENTO

09a-before-1920x1080-es-dark.png

BEFORE — 1920x1080 ES dark.

04 / HARDWARE

EXPERIENCIA DE HARDWARE

DENTRO DEL MONTAJE

Construido desde dentro.

Cada componente juzgado por lo que hace,
no por lo que promete la caja.

01 MONTAJE

La máquina como objeto acabado: montada, cableada y encendida.

02 COMPONENTES

Las piezas que de verdad importan, examinadas de cerca: GPU, CPU, refrigeración, almacenamiento.

03 RENDIMIENTO

No es una tabla de benchmarks: lo que el hardware permite de verdad cada día.

09b-after-1920x1080-es-dark.png

AFTER — 1920x1080 ES dark. Spanish is the longest headline of the three and still clears the object.

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not what the box promises.

01 BUILD

The machine as a finished object — assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined up close: GPU, CPU, cooling, storage.

03 PERFORMANCE

Not a benchmark chart — what the hardware actually enables day to day.

INSIDE THE BUILD

Built from the inside out.

Every component judged on what it does, not what the box promises.



01 BUILD

The machine as a finished object — assembled, cabled, and powered on.

02 COMPONENTS

The individual pieces that matter, examined up close: GPU, CPU, cooling, storage.

03 PERFORMANCE

Not a benchmark chart — what the hardware actually enables day to day.

Validation

Command	Result
npm run typecheck	pass, 0 errors
npm run lint	pass, 0 problems
npm run test:run	42/42
npm run build	pass
npm run test:e2e	69/69 (67 existing + 2 new)
git diff --check	clean

Four full e2e runs: 69/69, 68/69, 69/69, 69/69. The single failure was *light theme: the Footer copy meets AA contrast*, which is not caused by this change and not in this section: the footer links transition their colour on reveal, and the test samples `getComputedStyle` without waiting, catching `rgb(116,117,118)` (4.22:1) instead of the settled `rgb(95,96,98)` (5.76:1). Running that exact sequence six times against each build returns the identical in-flight colour from BOTH — the pre-change build fails it 6/6 as well. It is a latent bug in the Footer's own test, left alone because the Footer is explicitly out of scope.

New tests

the #hardware anchor lands below the fixed bar at every width, in every language — 4 viewports × 3 languages, asserts positive clearance. Waits for the smooth scroll to actually come to rest rather than sleeping a fixed interval.

Hardware: the object never crosses the copy, and the pin never strands the tail — 3 desktop viewports × 3 languages, asserts $0px^2$ of painted-ink overlap and that the section is no taller than the viewport it pins in.

Git status

```
M e2e/app.spec.ts
M src/sections/hardware/Hardware.module.css
M src/sections/hardware/Hardware.tsx

3 files changed, 262 insertions(+), 20 deletions(-)
baseline: d0b8ecf feat(featured): add editorial story progress markers
```

Nothing committed, pushed, reset, cleaned or reverted.

Verdict

Delivered. Hardware now presents as one composed frame: copy and object in their own columns with no collision in any language, three chapter markers as a band across the base, the whole section exactly one viewport at every standard desktop size, an anchor that lands below the bar at every width, an object that is legible as material instead of reading as a hole, and a pin that can no longer borrow scroll from a reader who has not finished the section.